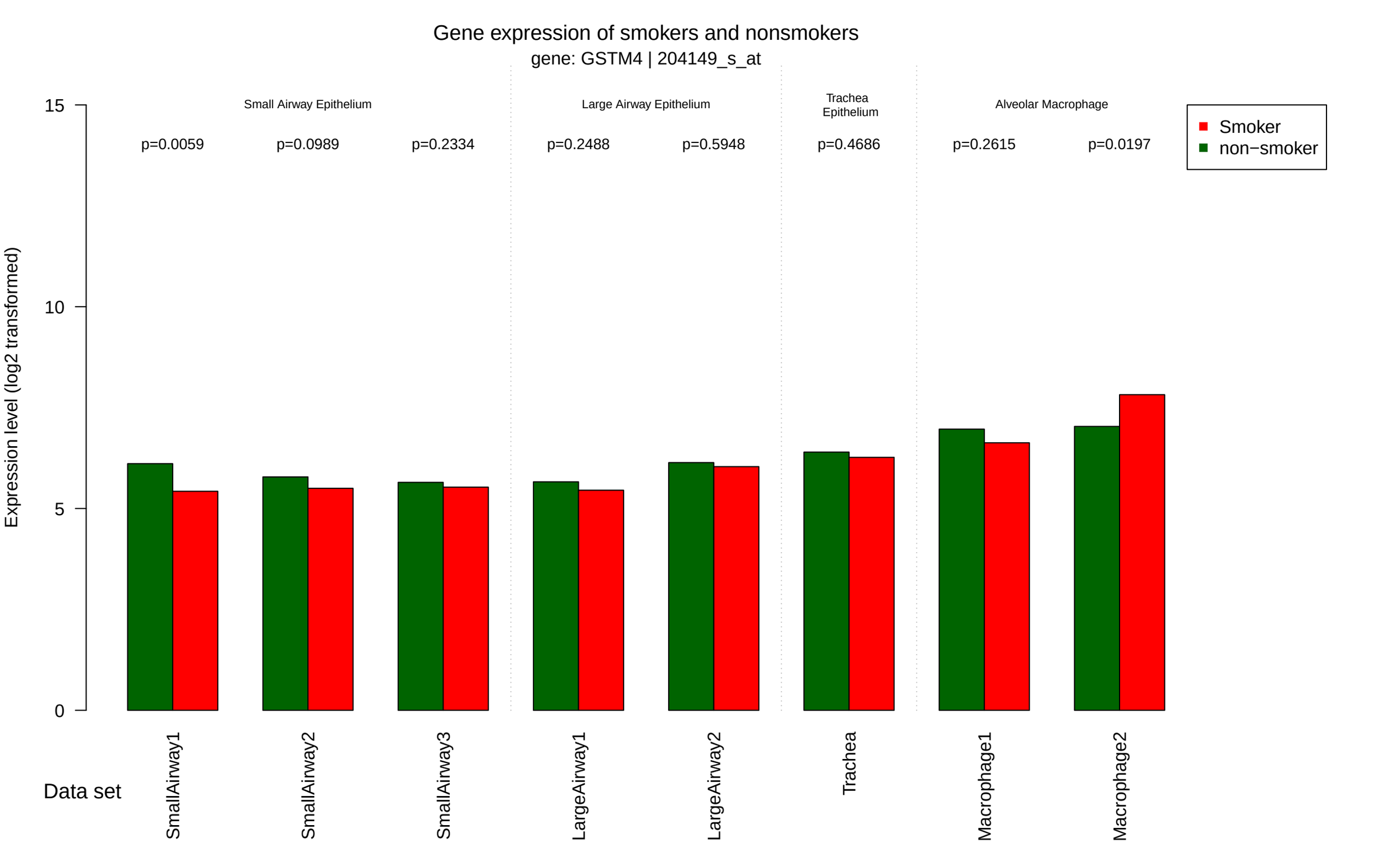
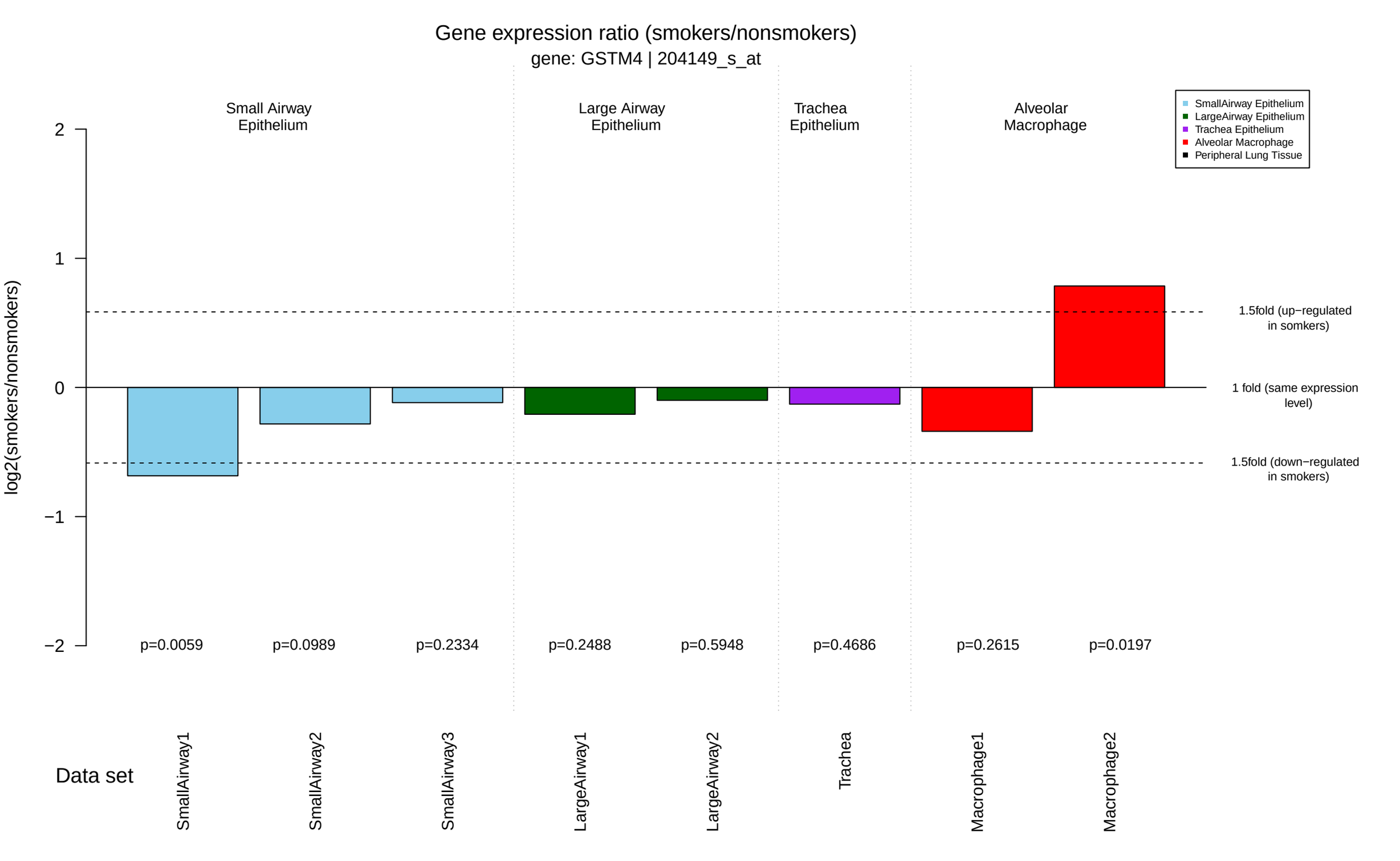
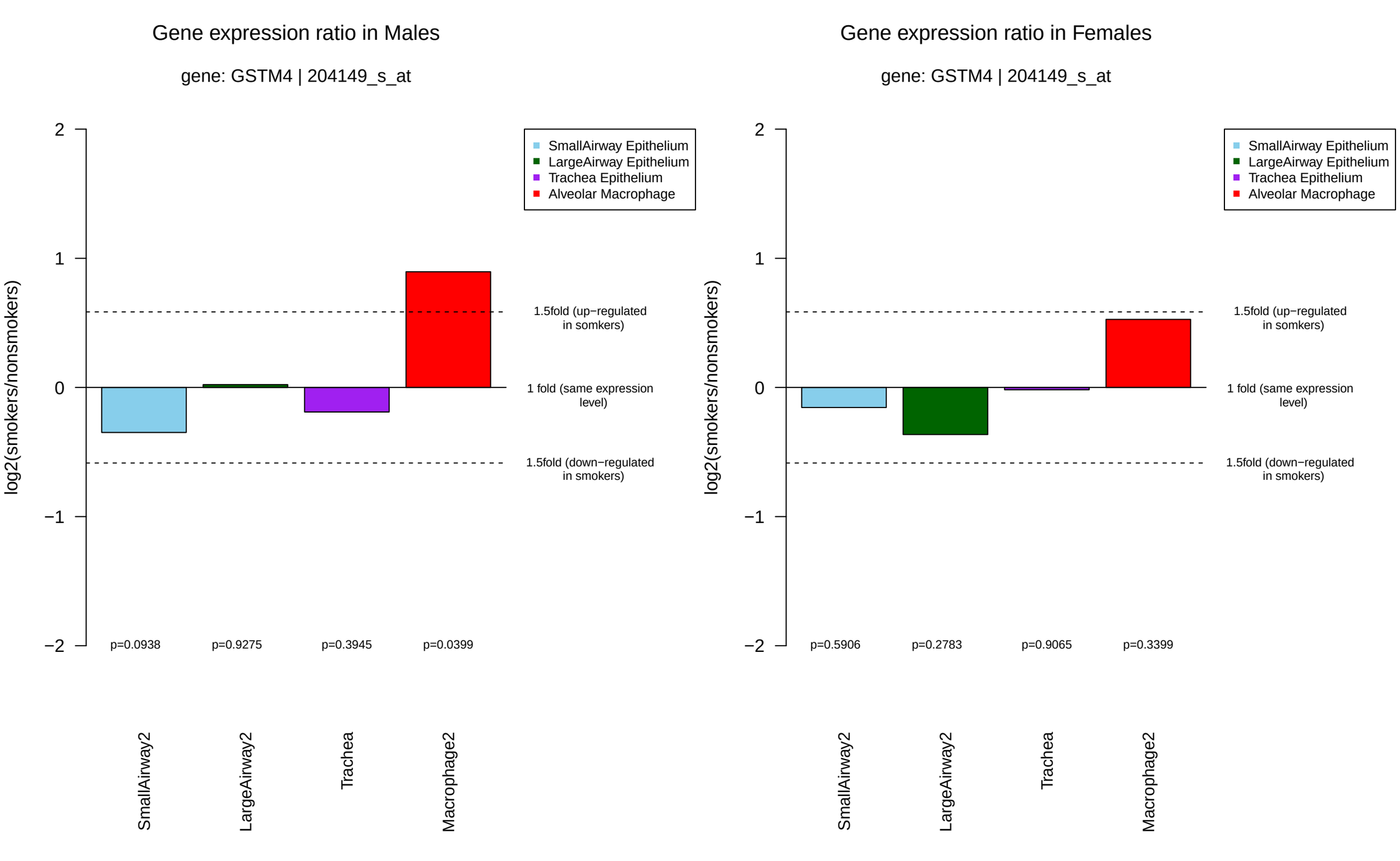
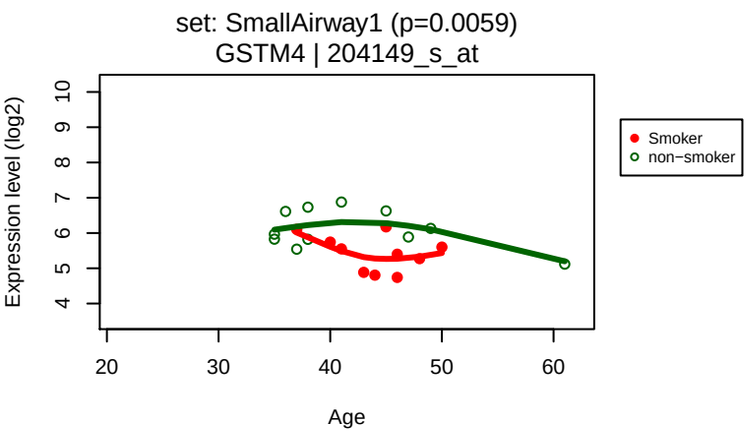
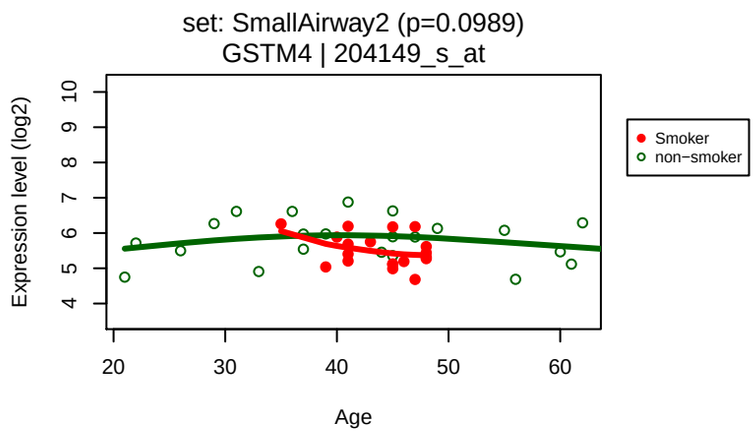
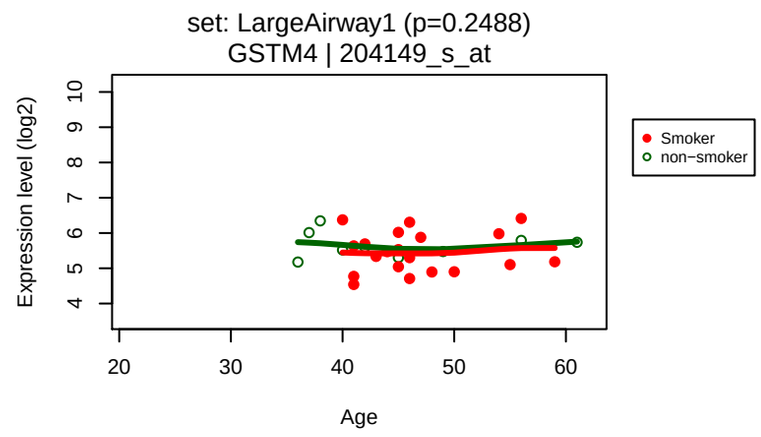

Gene expression of smokers and nonsmokers
gene: GSTM4 | 204149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0059
p=0.0989
p=0.2334
p=0.2488
p=0.5948
p=0.4686
p=0.2615
p=0.0197
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

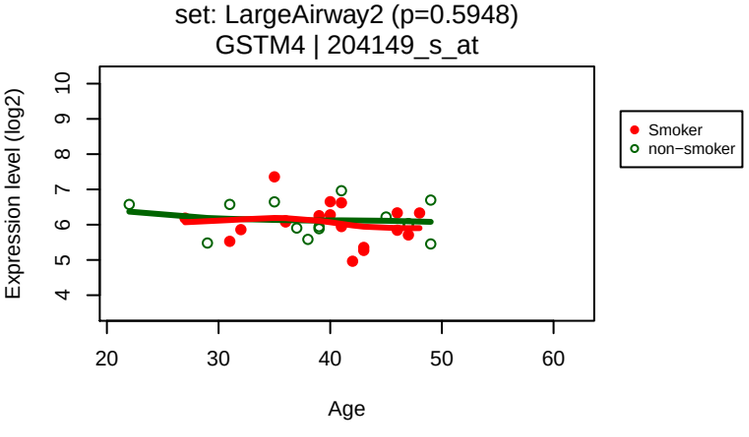

Gene expression ratio (smokers/nonsmokers)
gene: GSTM4 | 204149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0059
p=0.0989
p=0.2334
p=0.2488
p=0.5948
p=0.4686
p=0.2615
p=0.0197
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

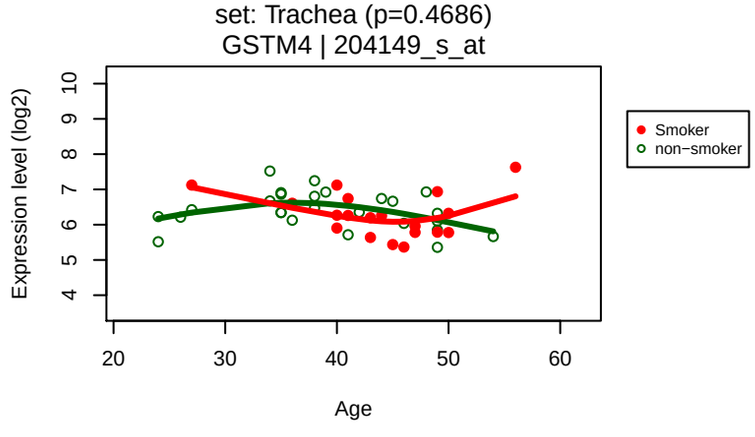

Gene expression ratio in Males
gene: GSTM4 | 204149_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0938
p=0.9275
p=0.3945
p=0.0399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTM4 | 204149_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5906
p=0.2783
p=0.9065
p=0.3399
SmallAirway2
LargeAirway2
Trachea
Macrophage2

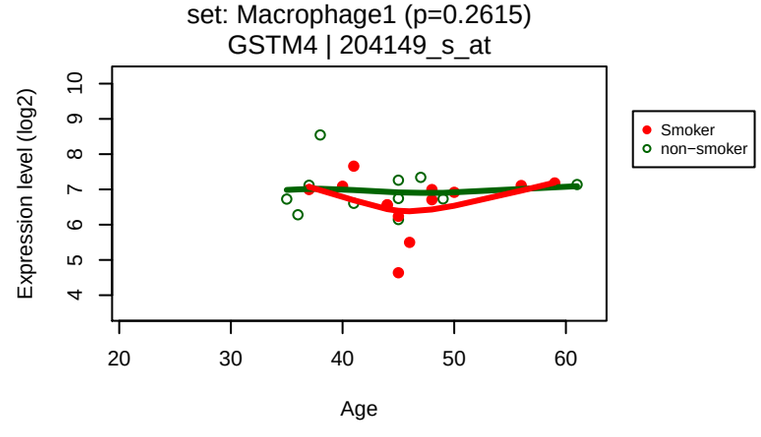

set: SmallAirway1 (p=0.0059)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker

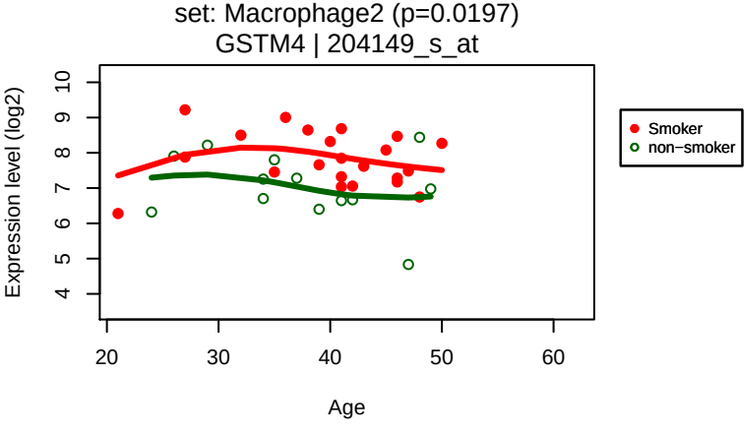

set: SmallAirway2 (p=0.0989)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker

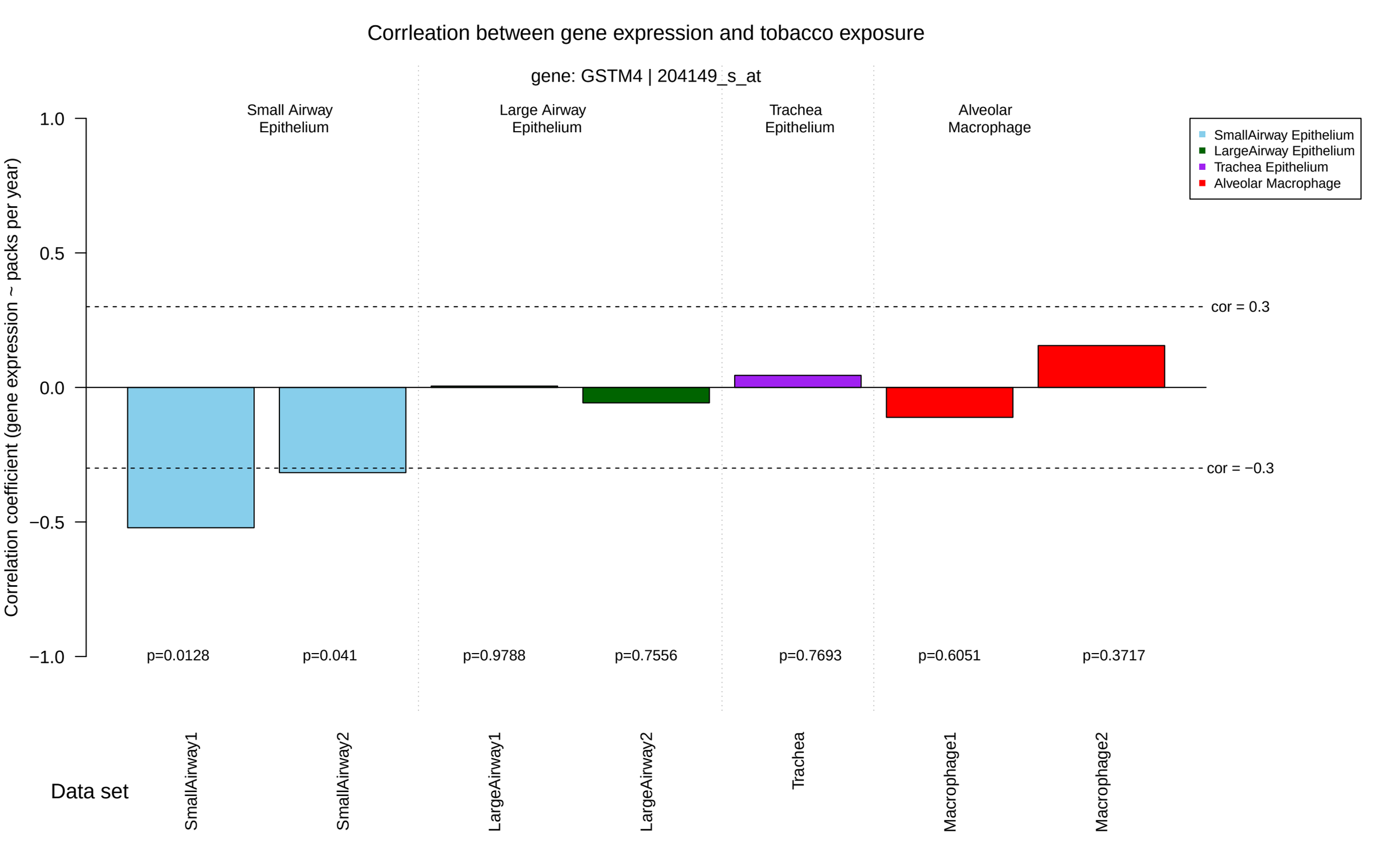

set: LargeAirway1 (p=0.2488)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5948)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4686)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2615)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0197)
GSTM4 | 204149_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTM4 | 204149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0128
p=0.041
p=0.9788
p=0.7556
p=0.7693
p=0.6051
p=0.3717
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2